

4.1 Consider the combinational circuit shown in Fig. P4.1. (HDL—see Problem 4.49.)

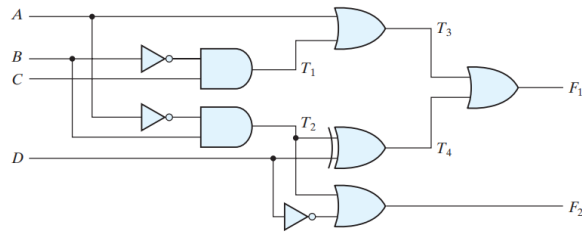


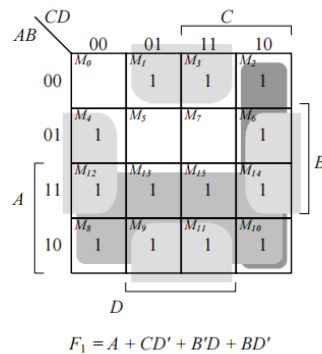
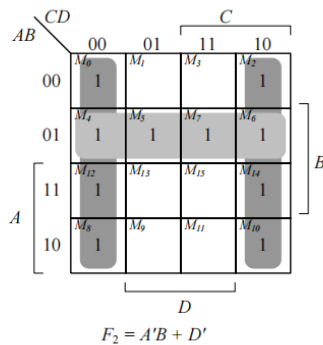
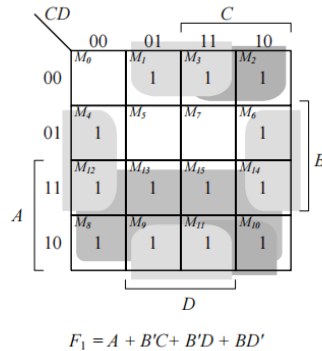
FIGURE P4.1

- (a)* Derive the Boolean expressions for T_1 through T_4 . Evaluate the outputs F_1 and F_2 as a function of the four inputs.
- (b) List the truth table with 16 binary combinations of the four input variables. Then list the binary values for T_1 through T_4 and outputs F_1 and F_2 in the table.
- (c) Plot the output Boolean functions obtained in part (b) on maps and show that the simplified Boolean expressions are equivalent to the ones obtained in part (a).

4.1

- (a) $T_1 = B'C$, $T_2 = A'B$, $T_3 = A + T_1 = A + B'C$,
 $T_4 = D \oplus T_2 = D \oplus (A'B) = A'BD' + D(A + B') = A'BD' + AD + B'D$
 $F_1 = T_3 + T_4 = A + B'C + A'BD' + AD + B'D$
 With $A + AD = A$ and $A + A'BD' = A + BD'$:
 $F_1 = A + B'C + BD' + B'D$
 Alternative cover: $F_1 = A + CD' + BD' + B'D$
 $F_2 = T_2 + D' = A'B + D'$

ABCD	T_1	T_2	T_3	T_4	F_1	F_2
0000	0	0	0	0	0	1
0001	0	0	0	1	1	0
0010	1	0	1	0	1	1
0011	1	0	1	1	1	0
0100	0	1	0	1	1	1
0101	0	1	0	0	0	1
0110	0	1	0	1	1	1
0111	0	1	0	0	0	1
1000	0	0	1	0	1	1
1001	0	0	1	1	1	0
1010	1	0	1	0	1	1
1011	1	0	1	1	1	0
1100	0	0	1	0	1	1
1101	0	0	1	1	1	0
1110	0	0	1	0	1	1
1111	0	0	1	1	1	0



4.9 An ABCD-to-seven-segment decoder is a combinational circuit that converts a decimal digit in BCD to an appropriate code for the selection of segments in an indicator used to display the decimal digit in a familiar form. The seven outputs of the decoder (a, b, c, d, e, f, g) select the corresponding segments in the display, as shown in Fig. P4.9(a). The numeric display chosen to represent the decimal digit is shown in Fig. P4.9(b). Using a truth table and Karnaugh maps, design the BCD-to-seven-segment decoder using a minimum number of gates. The six invalid combinations should result in a blank display. (HDL—see Problem 4.51.)



(a) Segment designation

(b) Numerical designation for display

$$\begin{aligned}
 a &= wx'y' + w'xz + w'y + \left\{ \begin{array}{l} x'y'z' \\ \text{or} \\ w'x'z' \end{array} \right\} \\
 b &= w'x' + w'yz + w'y'z' + x'y' \\
 c &= w'z + w'x + x'y' \\
 d &= wx'y' + w'x'y + w'xy'z + w'yz' + \left\{ \begin{array}{l} x'y'z' \\ \text{or} \\ w'x'z' \end{array} \right\} \\
 e &= x'y'z' + w'yz' + w'x'z' \\
 f &= wx'y' + w'xy' + w'xz' + \left\{ \begin{array}{l} x'y'z' \\ \text{or} \\ w'y'z' \end{array} \right\} \\
 g &= wx'y' + w'xy' + w'y'z' + w'x'y
 \end{aligned}$$

$$\begin{aligned}
 a &= AB'C' + A'BD + A'C + \left\{ \begin{array}{l} BCD \\ \text{or} \\ A'B'D' \end{array} \right\} \\
 b &= A'B' + A'CD + A'C'D' + B'C' \\
 c &= A'D + A'B + B'C' \\
 d &= AB'C' + A'B'C + A'BC'D + A'CD' + \left\{ \begin{array}{l} B'C'D' \\ \text{or} \\ A'B'D' \end{array} \right\} \\
 e &= B'C'D' + A'CD' + A'B'D' \\
 f &= AB'C' + A'BC' + A'BD + \left\{ \begin{array}{l} B'C'D' \\ \text{or} \\ A'C'D' \end{array} \right\} \\
 g &= AB'C' + A'BC' + A'CD' + A'B'C
 \end{aligned}$$

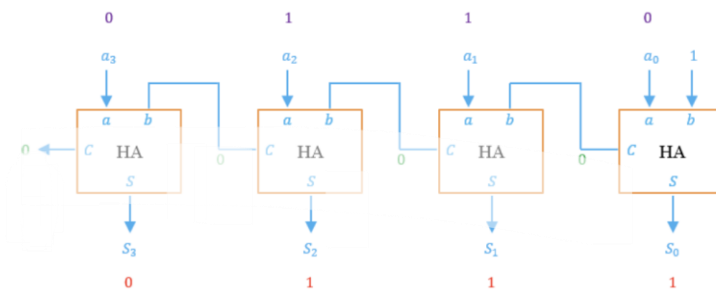
or

4.11 Using four half-adders (HDL—see Problem 4.52),

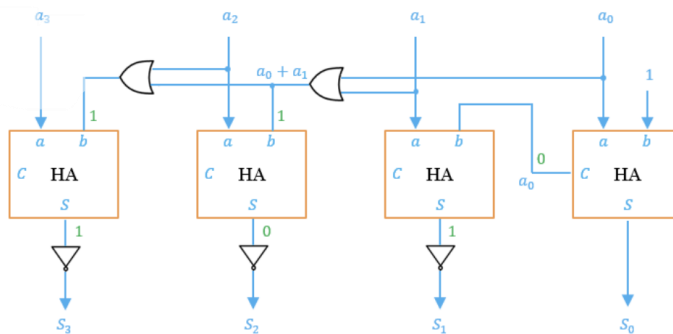
- (a) Design a full-subtractor circuit incrementer. (A circuit that adds one to a four-bit binary number.)
- (b)* Design a four-bit combinational decremter (a circuit that subtracts 1 from a four-bit binary number).

使用四个半加器（HDL——参见问题 4.52），(a) 设计一个全减器电路增量器（该电路可对四位二进制数进行加一操作）。(b)* 设计一个四位组合减计数器（该电路可对四位二进制数进行减 1 操作）

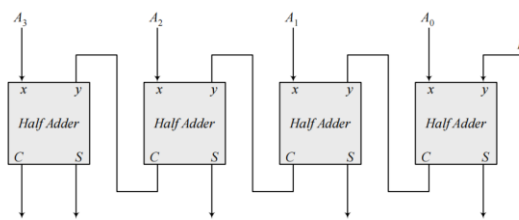
a



B

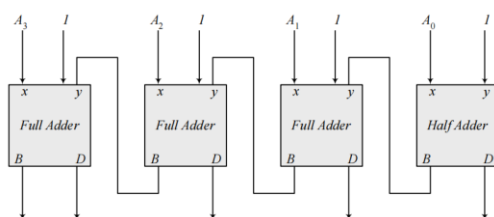


Or



Note: 5-bit output

b)



4.16 Define the carry propagate and carry generate as

$$P_i = A_i + B_i$$

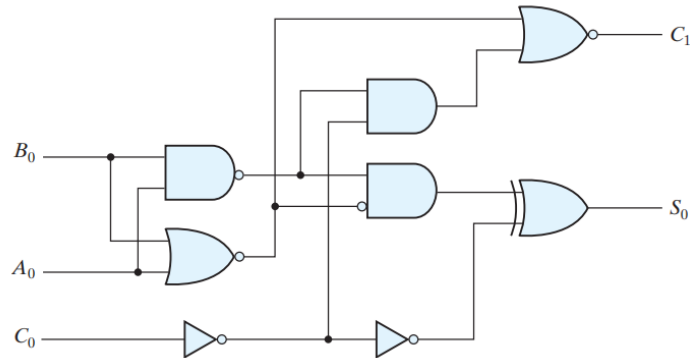
$$G_i = A_i B_i$$

respectively. Show that the output carry and output sum of a full adder becomes

$$C_{i+1} = (C_i' G_i' + P_i)'$$

$$S_i = (P_i G'_i) \oplus C_i$$

The logic diagram of the first stage of a four-bit parallel adder as implemented in IC type 74283 is shown in Fig. P4.16. Identify the P'_i and G'_i terminals and show that the circuit implements a full-adder circuit.



(a)

$$\begin{aligned} (C'G'_i + p'_i)' &= (C_i + G_i)P_i = G_iP_i + P_iC_i \\ &= A_iB_i(A_i + B_i) + P_iC_i \\ &= A_iB_i + P_iC_i = G_i + P_iC_i \\ &= A_iB_i + (A_i + B_i)C_i = A_iB_i + A_iC_i + B_iC_i = C_{i+1} \\ (P_iG'_i) \oplus C_i &= (A_i + B_i)(A_iB_i)' \oplus C_i = (A_i + B_i)(A'_i + B'_i) \oplus C_i \\ &= (A'_iB_i + A_iB'_i) \oplus C_i = A_i \oplus B_i \oplus C_i = S_i \end{aligned}$$

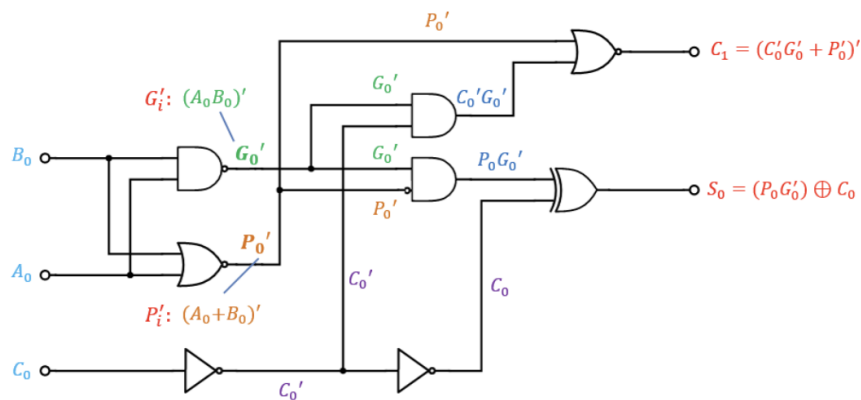
(b)

Output of NOR gate = $(A_0 + B_0)' = P'_0$

Output of NAND gate = $(A_0B_0)' = G'_0$

$$S_1 = (P_0 G'_0) \oplus C_0$$

$$C_1 = (C'_0 G'_0 + P'_0)' \quad \text{as defined in part (a)}$$



4.17 Show that the output carry in a full adder circuit can be expressed in the AND-OR-INVERT form

$$C_{i+1} = G_i + P_i C_i = (G'_i P'_i + G'_i C'_i)'$$

IC type 74182 is a lookahead carry generator circuit that generates the carries with AND-OR-INVERT gates (see Section 3.8). The circuit assumes that the input terminals have the complements of the G 's, the P 's, and of C_1 . Derive the Boolean functions for the lookahead carries C_2 , C_3 , and C_4 in this IC. (*Hint*: Use the equation-substitution method to derive the carries in terms of C'_i)

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$$\begin{aligned} C_{i+1} &= G_i + P_i C_i \\ &= ((G_i + P_i C_i)')' \Rightarrow (x')' = x \\ &= ((G_i)'(P_i C_i)')' \Rightarrow \text{(DeMorgan)} \\ &= (G'_i(P'_i + C'_i))' \Rightarrow \text{(DeMorgan)} \\ &= \boxed{(G'_i P'_i + G'_i C'_i)'} \end{aligned}$$

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$$C_2 = \boxed{(G'_1 P'_1 + G'_1 C'_1)'}$$

$$\begin{aligned} C_3 &= (G'_2 P'_2 + G'_2 C'_2)' \\ &= (G'_2 P'_2 + G'_2(G'_1 P'_1 + G'_1 C'_1))' \Rightarrow \text{(From Eq.1)} \\ &= \boxed{(G'_2 P'_2 + G'_2 G'_1 P'_1 + G'_2 G'_1 C'_1)'} \end{aligned}$$

$$\begin{aligned} C_4 &= (G'_3 P'_3 + G'_3 C'_3)' \\ &= (G'_3 P'_3 + G'_3(G'_2 P'_2 + G'_2 G'_1 P'_1 + G'_2 G'_1 C'_1))' \Rightarrow \text{(From Eq.2)} \\ &= \boxed{(G'_3 P'_3 + G'_3 G'_2 P'_2 + G'_3 G'_2 G'_1 P'_1 + G'_3 G'_2 G'_1 C'_1)'} \end{aligned}$$

$$\begin{aligned} C_2 &= (G'_1 P'_1 + G'_1 C'_1)' \\ C_3 &= (G'_2 P'_2 + G'_2 G'_1 P'_1 + G'_2 G'_1 C'_1)' \\ C_4 &= (G'_3 P'_3 + G'_3 G'_2 P'_2 + G'_3 G'_2 G'_1 P'_1 + G'_3 G'_2 G'_1 C'_1)' \end{aligned}$$

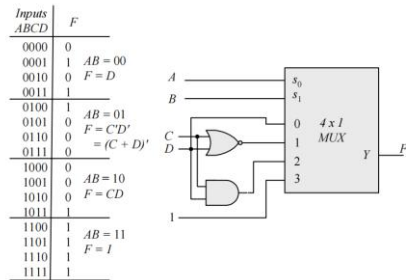
4.35 Implement the following Boolean function with a 4×1 multiplexer and external gates.

(a) * $F_1(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$

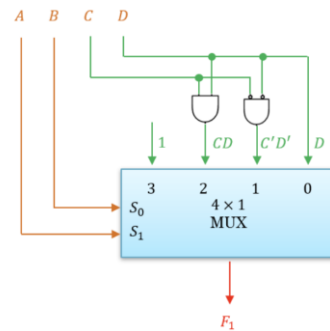
(b) $F_2(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$

Connect inputs A and B to the selection lines. The input requirements for the four data lines will be a function of variables C and D . These values are obtained by expressing F as a function of C and D for each of the four cases when $AB = 00, 01, 10$, and 11 . These functions may have to be implemented with external gates. (HDL—see Problem 4.47.)

(a)

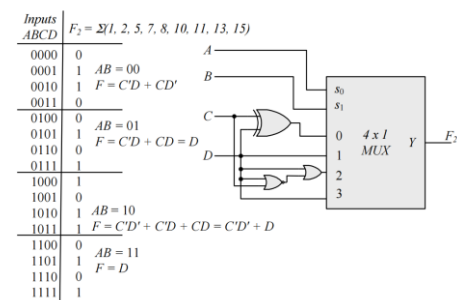


or



(b)

$F = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$



or

